1 Results of the synthetic time series analyses

Synthetic time series, generated using equation 13 with random seed 94. Univariate analysis identified an ARIMA$(2, 1, 1)$ model; based on this, tables 1, 2, and 3 present the results for the ANFIS modeling.

Table 1: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 94, with 1 lag.

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,0003420	0,0184931	0,0134072	0,2184308	93,95
$DB(t-1)+Res$	0,0102946	0,1014625	0,0885179	1,4444388	77,96
$DB(t-1)+Res(t-1)$	0,0064317	0,0801981	0,0577827	0,9451070	123,03
$DB(t-1)+Dif$	0,0000265	0,0051223	0,0048101	0,0783685	493,99
$DB(t-1)+Dif(t-1)$	0,0013831	0,0371897	0,0309245	0,5054367	132,47
$DB(t-1)+Res+Dif$	0,0006513	0,0255205	0,0242613	0,3955757	172,61
$DB(t-1)+Res(t-1)+Dif$	0,0006528	0,0255499	0,0239079	0,3902191	394,64
$DB(t-1)+Res+Dif(t-1)$	0,0078573	0,0886414	0,0745758	1,2193039	155,69
$DB(t-1)+Res(t-1)+Dif(t-1)$	0,0058098	0,0762221	0,0600138	0,9810489	217,67

Source: Prepared by the author

Table 2: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 94, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,0018884	0,0434553	0,0395147	0,6446143	186,50
$DB(t-1)+DB(t-2)+Res$	0,0028620	0,0534978	0,0492357	0,8033182	106,63
$DB(t-1)+DB(t-2)+Res(t-1)$	0,0019710	0,0443962	0,0375470	0,6131414	237,06
$DB(t-1)+DB(t-2)+Dif$	0,0003599	0,0189707	0,0130730	0,2137181	260,66
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0017747	0,0421269	0,0275191	0,4498536	272,29
$DB(t-1)+DB(t-2)+Res+Dif$	0,0014339	0,0378669	0,0255381	0,4171558	772,38
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0081626	0,0903473	0,0681590	1,1139412	814,52
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	0,0104650	0,1022988	0,0665167	1,0855770	217,43
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,0097299	0,0986402	0,0692816	1,1331249	285,87

Source: Prepared by the author

Table 3: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 94, with 3 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1) + DB(t-2) + DB(t-3)$	0,0006526	0,0255462	0,0210872	0,3443080	228,45
$DB(t-1) + DB(t-2) + DB(t-3) + Res$	0,0075024	0,0866162	0,0599611	0,9787914	216,31
$DB(t-1) + DB(t-2) + DB(t-3) + Res(t-1)$	0,0349082	0,1868373	0,1159764	1,8983475	328,34
$DB(t-1) + DB(t-2) + DB(t-3) + Dif$	0,0024977	0,0499773	0,0460392	0,7508351	679,10
$DB(t-1) + DB(t-2) + DB(t-3) + Dif(t-1)$	0,0195390	0,1397819	0,1140421	1,8622916	286,93
$DB(t-1) + DB(t-2) + DB(t-3) + Res + Dif$	0,0133814	0,1156782	0,0672449	1,0984120	352,17
$DB(t-1) + DB(t-2) + DB(t-3)+Res(t-1)+Dif$	0,0041990	0,0647995	0,0565671	0,9232327	681,22
$DB(t-1) + DB(t-2) + DB(t-3)+Res+Dif(t-1)$	0,0086307	0,0929017	0,0770218	1,2530785	437,50
$DB(t-1) + DB(t-2) + DB(t-3) + Res(t-1) + Dif(t-1)$	0,0747681	0,2734375	0,1867452	3,0532103	395,16

Source: Prepared by the author

Tables 4 and 5 present the results for the ensemble of selected models and a comparison of the best models, respectively.

Table 4: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 94

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,000508763	0,022555766	0,01860905	0,304079
Weighted average ($60\%ARIMA$) ($40\%ANFIS$)	0,000695690	0,026375943	0,021588011	0,352787
Weighted average ($40\%ARIMA$) ($60\%ANFIS$)	0,000351976	0,018761018	0,015630089	0,255371
Weighted average ($80\%ARIMA$) ($20\%ANFIS$)	0,001159969	0,034058317	0,02762982	0,451559
Weighted average ($20\%ARIMA$) ($80\%ANFIS$)	0,000128825	0,011350116	0,010106874	0,165028
Weighted average using $RMSE$	0,000067336	0,008205856	0,007703535	0,125708

Source: Prepared by the author

Table 5: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 94

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(2, 1, 1)$	0,0017448	0,0417709	0,0342282	0,5639812
ANFIS $DB(t-1) + Dif$	0,0000265	0,0051223	0,0048101	0,0783685
ANFIS $DB(t-1)+DB(t-2)+Dif$	0,0003599	0,0189707	0,0130730	0,2137181
ANFIS $DB(t-1) + DB(t-2) + DB(t-3)$	0,0006526	0,0255462	0,0210872	0,3443080
Weighted average using $RMSE$	0,0000673	0,0082059	0,0077035	0,1257080

Source: Prepared by the author

Synthetic time series, generated using equation 13, with random seed number 95. Univariate analysis identified an ARIMA$(1, 1, 1)$ model; based on this, tables 6 and 7 present the results for ANFIS modeling.

Table 6: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 95, with 1 lag

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,0085327	0,0923727	0,0848468	1,3816037	94,07
$DB(t-1) + Res$	0,0045093	0,0671513	0,0565098	0,6681126	128,04
$DB(t-1) + Res(t-1)$	0,7465324	0,08640211	0,3548089	5,7387076	187,07
$DB(t-1) + Dif$	0,0000712	0,0084399	0,0081481	0,0964750	1049,01
$DB(t-1) + Dif(t-1)$	0,0105451	0,1026894	0,0905957	1,4738598	137,84
$DB(t-1) + Res + Dif$	0,0663928	0,2576680	0,171060	2,778702	111,19
$DB(t-1)+Res(t-1)+Dif$	0,0053913	0,0734255	0,0300943	0,3520901	1260,83
$DB(t-1) + Res + Dif(t-1)$	0,0089808	0,0947672	0,0750198	1,2232621	126,67
$DB(t-1)+Res(t-1)+Dif(t-1)$	0,0537543	0,2318496	0,1476507	2,3969124	166,10

Source: Prepared by the author

Table 7: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 95, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,0010283	0,0320668	0,0283496	0,3375118	196,46
$DB(t-1)+DB(t-2)+Res$	0,0067043	0,0818798	0,0698556	0,8270858	1778,71
$DB(t-1)+DB(t-2)+Res(t-1)$	0,0180069	0,134190	0,1118799	1,3240243	248,26
$DB(t-1)+DB(t-2)+Dif$	0,0019432	0,0440818	0,0276079	0,3272233	362,53
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0102618	0,1013007	0,0894735	1,4555838	107,04
$DB(t-1)+DB(t-2)+Res+Dif$	0,1245125	0,3528633	0,2395377	3,8894171	203,44
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	1,3940848	1,1807137	0,5071532	5,9362346	283,75
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	0,0439552	0,2096551	0,1396914	2,268078	172,03
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,0262195	0,1619242	0,1304058	2,1215168	192,02

Source: Prepared by the author

Tables 8 and 9 present the results for the ensemble of selected models and a comparison of the best models, respectively.

Table 8: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 95

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,001061182	0,032575782	0,027679070	0,326885
Weighted average $(60\%ARIMA)$ $(40\%ANFIS)$	0,001616557	0,040206435	0,0338388	0,399526
Weighted average $(40\%ARIMA)$ $(60\%ANFIS)$	0,000624935	0,024998692	0,021519343	0,254244
Weighted average $(80\%ARIMA)$ $(20\%ANFIS)$	0,003084695	0,055540034	0,046158257	0,544809
Weighted average $(20\%ARIMA)$ $(80\%ANFIS)$	0,000109827	0,010479833	0,009199886	0,108962
Weighted average using $RMSE$	0,000032432	0,005694951	0,005085608	0,060534

Source: Prepared by the author

Table 9: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 95

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(1,1,1)$	0,0050293	0,0709178	0,0584775	0,6839985
ANFIS $DB(t-1)+Dif$	0,0000712	0,0084399	0,0081481	0,0964750
ANFIS $DB(t-1)+DB(t-2)$	0,0010283	0,0320668	0,0283496	0,3375118
Weighted average using $RMSE$	0,0000324	0,0056949	0,0050856	0,0605340

Source: Prepared by the author

Synthetic time series, generated using equation 13 with random seed 155. Univariate analysis identified an ARIMA$(1,1,1)$ model; based on this, tables 10 and 11 present the results for ANFIS modeling.

Table 10: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 155, with 1 lag

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,0021664	0,0465444	0,0448879	0,825211	67,51
$DB(t-1)+Res$	0,0073230	0,0855744	0,0815483	1,5024649	89,35
$DB(t-1)+Res(t-1)$	0,0025427	0,0504254	0,8903208	0,8903208	286,89
$DB(t-1)+Dif$	0,0000187	0,0043214	0,0040582	0,0742847	240,83
$DB(t-1)+Dif(t-1)$	0,0036728	0,0606039	0,0576450	1,0599242	95,06
$DB(t-1)+Res+Dif$	0,0002552	0,0159742	0,0153053	0,2813626	325,31
$DB(t-1)+Res(t-1)+Dif$	0,0001187	0,0108971	0,0098977	0,1808103	485,77
$DB(t-1)+Res+Dif(t-1)$	0,0016086	0,0401077	0,0353553	0,6541974	126,82
$DB(t-1)+Res(t-1)+Dif(t-1)$	0,0059880	0,0773821	0,0551036	1,0131488	186,00

Source: Prepared by the author

Table 11: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 155, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,0033115	0,0575460	0,0520817	0,9589879	99,24
$DB(t-1)+DB(t-2)+Res$	0,0106331	0,1031171	0,0936890	1,7137501	212,19
$DB(t-1)+DB(t-2)+Res(t-1)$	0,0028118	0,0530265	0,0503095	0,9227892	207,32
$DB(t-1)+DB(t-2)+Dif$	0,0003158	0,0177716	0,0161185	0,2964414	246,28
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0012459	0,0352975	0,0327064	0,6037818	124,75
$DB(t-1)+DB(t-2)+Res+Dif$	0,0007981	0,0282513	0,0187321	0,3486456	327,36
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0002418	0,0155488	0,0107175	0,1996996	683,38
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	0,0009993	0,0316114	0,0269720	0,4964091	167,34
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,0054594	0,0738878	0,0665785	1,2300533	283,32

Source: Prepared by the author

Tables 12 and 13 present the results for the ensemble of selected models and a comparison of the best models, respectively.

Table 12: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 155

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,000108926	0,010436764	0,007578293	0,141461
Weighted average $(60\%ARIMA)$ $(40\%ANFIS)$	0,000167350	0,012936393	0,009442463	0,176113
Weighted average $(40\%ARIMA)$ $(60\%ANFIS)$	0,000063960	0,007997485	0,005766626	0,107775
Weighted average $(80\%ARIMA)$ $(20\%ANFIS)$	0,000324573	0,018015897	0,013229717	0,246491
Weighted average $(20\%ARIMA)$ $(80\%ANFIS)$	0,000014401	0,003794872	0,003212251	0,059786
Weighted average using $RMSE$	0,000010386	0,003222696	0,003080502	0,056918

Source: Prepared by the author

Table 13: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 155

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(1, 1, 1)$	0,0005356	0,02314363	0,01731722	0,3204714
ANFIS $DB(t-1) + Dif$	0,0000187	0,0043214	0,0040582	0,0742847
ANFIS $DB(t-1)+DB(t-2)+Dif$	0,0003158	0,0177716	0,0161185	0,2964414
Weighted average using $RMSE$	0,0000104	0,0032227	0,0030805	0,056918

Source: Prepared by the author

Synthetic time series, generated using equation 13 with random seed 185. Univariate analysis identified an ARIMA$(2, 1, 1)$ model; based on this, tables 14, 15, and 16 present the results for the ANFIS modeling.

Table 14: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 185, with 1 lag

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,002303	0,0479891	0,0396704	0,7328429	86,08
$DB(t-1) + Res$	0,0312660	0,1768218	0,1485925	2,7511471	206,59
$DB(t-1) + Res(t-1)$	0,0045785	0,0676649	0,0623185	1,1418961	320,19
$DB(t-1) + Dif$	0,0000470	0,0068578	0,0062866	0,1152640	605,89
$DB(t-1) + Dif(t-1)$	0,0033847	0,0581778	0,0468997	0,8653491	114,58
$DB(t-1) + Res + Dif$	0,0336615	0,1834707	0,0842664	1,5753252	405,77
$DB(t-1)+Res(t-1)+Dif$	0,0004360	0,0208806	0,0190171	0,3512800	421,13
$DB(t-1) + Res + Dif(t-1)$	0,0038171	0,0617826	0,0517553	0,9543319	191,33
$DB(t-1)+Res(t-1)+Dif(t-1)$	0,9377450	0,9683724	0,4853447	8,9919792	165,96

Source: Prepared by the author

Table 15: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 185, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,0067475	0,0821434	0,0706845	1,309558	124,66
$DB(t-1)+DB(t-2)+Res$	0,0656024	0,2561297	0,1304949	2,4366608	228,98
$DB(t-1)+DB(t-2)+Res(t-1)$	0,1357394	0,3684283	0,2580352	4,7914864	189,31
$DB(t-1)+DB(t-2)+Dif$	0,0003679	0,0191811	0,0173773	0,3208531	577,04
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0309934	0,1760495	0,1651751	3,0260109	273,41
$DB(t-1)+DB(t-2)+Res+Dif$	0,0004148	0,0203670	0,0165469	0,3066090	389,70
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0591660	0,2432406	0,1687055	3,1390066	588,05
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	0,0200964	0,1417616	0,1255918	2,1255918	324,44
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,0560821	0,2368165	0,1952388	3,6118432	378,45

Source: Prepared by the author

Table 16: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 185, with 3 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)+DB(t-3)$	0,0187135	0,1367973	0,1246382	2,2852269	156,12
$DB(t-1)+DB(t-2)+DB(t-3)+Res$	0,0224064	0,1496876	0,1318631	2,4226088	142,73
$DB(t-1)+DB(t-2)+DB(t-3)+Res(t-1)$	0,0317189	0,1780980	0,1611593	2,9613446	137,40
$DB(t-1)+DB(t-2)+DB(t-3)+Dif$	0,0032969	0,0574190	0,0387358	0,7194933	358,57
$DB(t-1)+DB(t-2)+DB(t-3)+Dif(t-1)$	0,0070822	0,0841558	0,0745788	1,3696931	275,95
$DB(t-1)+DB(t-2)+DB(t-3)+Res+Dif$	0,0731411	0,2704462	0,2248163	4,1716598	250,81
$DB(t-1)+DB(t-2)+DB(t-3)+Res(t-1)+Dif$	0,0208453	0,1443790	0,1063925	1,9725403	3763,72
$DB(t-1)+DB(t-2)+DB(t-3)+Res+Dif(t-1)$	0,0079792	0,0893262	0,0668849	1,2360511	237,95
$DB(t-1)+DB(t-2)+DB(t-3)+Res(t-1)+Dif(t-1)$	0,3972976	0,6303155	0,5338232	9,9083013	210,93

Source: Prepared by the author

Tables 17 and 18 present the results for the ensemble of selected models and a comparison of the best models, respectively.

Table 17: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 185

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,000809645	0,028454259	0,025100750	0,464888
Weighted average (60%$ARIMA$) (40%$ANFIS$)	0,001179074	0,034337642	0,030848514	0,570957
Weighted average (40%$ARIMA$) (60%$ANFIS$)	0,000512518	0,022638859	0,019475494	0,361043
Weighted average (80%$ARIMA$) (20%$ANFIS$)	0,002134837	0,046204294	0,042344043	0,783094
Weighted average (20%$ARIMA$) (80%$ANFIS$)	0,000135170	0,011626243	0,009322897	0,173280
Weighted average using $RMSE$	0,000057488	0,007582102	0,006096508	0,113034

Source: Prepared by the author

Table 18: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 185

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(2, 1, 1)$	0,0033798	0,0581361	0,0538396	0,9837380
ANFIS $DB(t-1) + Dif$	0,0000470	0,0068578	0,0062866	0,1152640
ANFIS $DB(t-1)+DB(t-2)+Res+Dif$	0,0003679	0,0191811	0,0173773	0,3208531
ANFIS $DB(t-1) + DB(t-2) + DB(t-3) + Dif$	0,0032969	0,0574190	0,0387358	0,7194933
Weighted average using $RMSE$	0,0000575	0,0075821	0,0060965	0,1130340

Source: Prepared by the author

Synthetic time series, generated using equation 13 with random seed 201. Univariate analysis identified the ARIMA$(1, 1, 1)$ model; based on this, tables 19 and 20 present the results for ANFIS modeling.

Table 19: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 201, with 1 lag

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,0008245	0,0287146	0,0228280	0,3518813	381,44
$DB(t-1) + Res$	0,1157070	0,3401573	0,2515738	3,8147541	128,18
$DB(t-1) + Res(t-1)$	0,2806925	0,5298042	0,3245907	4,9069996	193,88
$DB(t-1) + Dif$	0,0001105	0,0105117	0,0103940	0,1590624	382,87
$DB(t-1) + Dif(t-1)$	0,0054803	0,0740290	0,0665312	1,0196012	127,84
$DB(t-1) + Res + Dif$	0,0145495	0,1206214	0,0887023	1,3441488	179,02
$DB(t-1)+Res(t-1)+Dif$	0,0064888	0,0805533	0,0616680	0,9362418	402,90
$DB(t-1) + Res + Dif(t-1)$	0,0389989	0,1974813	0,1413714	2,1616533	156,59
$DB(t-1)+Res(t-1)+Dif(t-1)$	0,1348684	0,3672443	0,2162204	3,2807583	328,81

Source: Prepared by the author

Table 20: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 201, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,0013227	0,0363694	0,0342364	0,5230371	234,02
$DB(t-1)+DB(t-2)+Res$	0,5423985	0,7364771	0,5251523	7,9534984	183,17
$DB(t-1)+DB(t-2)+Res(t-1)$	0,1270984	0,3565086	0,2830010	4,2956802	217,44
$DB(t-1)+DB(t-2)+Dif$	0,0007962	0,0282173	0,0212893	0,3245726	334,44
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0126208	0,1123425	0,0979250	1,4889674	175,38
$DB(t-1)+DB(t-2)+Res+Dif$	0,0203820	0,1427654	0,1037769	1,5713655	1335,94
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0259788	0,1611794	0,0997803	1,5090494	691,82
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	0,0227918	0,1509695	0,1099616	1,6760636	304,82
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,4821025	0,6943360	0,4205705	6,3531757	305,99

Source: Prepared by the author

Tables 21 and 22 present the results for the ensemble of selected models and a comparison of the best models, respectively.

Table 21: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 201

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,002605164	0,051040808	0,039930657	0,605755
Weighted average $(60\%ARIMA)$ $(40\%ANFIS)$	0,003926627	0,062662801	0,048700126	0,738572
Weighted average $(40\%ARIMA)$ $(60\%ANFIS)$	0,001557878	0,039469963	0,031161189	0,472937
Weighted average $(80\%ARIMA)$ $(20\%ANFIS)$	0,007392081	0,085977212	0,066239063	1,004206
Weighted average $(20\%ARIMA)$ $(80\%ANFIS)$	0,000285835	0,016906656	0,013622251	0,207303
Weighted average using $RMSE$	0,000052374	0,007236977	0,006298804	0,096763

Source: Prepared by the author

Table 22: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 201

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(1,1,1)$	0,0119543	0,1093356	0,0837782	1,2433090
ANFIS $DB(t-1)+Dif$	0,0001105	0,0105117	0,0103940	0,1590624
ANFIS $DB(t-1)+DB(t-2)+Dif$	0,0007962	0,0282173	0,0212893	0,3245726
Weighted average using $RMSE$	0,0000524	0,0072370	0,0062988	0,0967630

Source: Prepared by the author

Synthetic time series, generated using equation 13 with random seed 312. Univariate analysis identified the ARIMA$(1,1,1)$ model; based on this, tables 23 and 24 present the results for ANFIS modeling.

Table 23: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 312, with 1 lag

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,0008155	0,0285573	0,0216435	0,3868932	86,04
$DB(t-1)+Res$	0,0004330	0,0208081	0,0177548	0,3180298	117,35
$DB(t-1)+Res(t-1)$	0,0004226	0,0205580	0,0143499	0,2588650	133,22
$DB(t-1)+Dif$	0,0003468	0,0186233	0,0119922	0,2140882	146,47
$DB(t-1)+Dif(t-1)$	0,0022392	0,0473206	0,0355693	0,6371244	122,85
$DB(t-1)+Res+Dif$	0,0001591	0,0126125	0,0102497	0,1830614	514,36
$DB(t-1)+Res(t-1)+Dif$	0,0001047	0,0102299	0,0084086	0,1500788	507,78
$DB(t-1)+Res+Dif(t-1)$	0,0012521	0,0353845	0,0301945	0,5432811	196,69
$DB(t-1)+Res(t-1)+Dif(t-1)$	0,0063585	0,0797400	0,0638962	1,1529002	176,05

Source: Prepared by the author

Table 24: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 312, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,0004937	0,0222186	0,0213329	0,3828106	293,27
$DB(t-1)+DB(t-2)+Res$	0,0030436	0,0551688	0,0496690	0,8897537	294,68
$DB(t-1)+DB(t-2)+Res(t-1)$	0,0043848	0,0662178	0,0476427	0,8491828	399,78
$DB(t-1)+DB(t-2)+Dif$	0,0006453	0,0254029	0,0225210	0,4038618	173,20
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0016786	0,0409710	0,0375224	0,6724667	171,00
$DB(t-1)+DB(t-2)+Res+Dif$	0,0063782	0,0798635	0,0670926	1,2077210	681,38
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0003164	0,0177888	0,0162739	0,2925936	453,20
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	0,0072819	0,0853343	0,0815492	1,4653883	293,51
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,0211272	0,1453520	0,1074676	1,9163933	471,94

Source: Prepared by the author

Tables 25 and 26 present the results for the ensemble of selected models and a comparison of the best models, respectively.

Table 25: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 312

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,000160634	0,012674149	0,011366664	0,204043
Weighted average $(60\%ARIMA)$ $(40\%ANFIS)$	0,000264500	0,016263457	0,014542874	0,260790
Weighted average $(40\%ARIMA)$ $(60\%ANFIS)$	0,000087658	0,009362579	0,008190454	0,147296
Weighted average $(80\%ARIMA)$ $(20\%ANFIS)$	0,000564901	0,023767655	0,020895294	0,374283
Weighted average $(20\%ARIMA)$ $(80\%ANFIS)$	0,000034375	0,005863022	0,004869771	0,087419
Weighted average using $RMSE$	0,000035636	0,005969570	0,005113353	0,091916

Source: Prepared by the author

Table 26: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 312

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(1, 1, 1)$	0,00098886	0,0314462	0,0272476	0,4858430
ANFIS $DB(t-1)+Res(t-1)+Dif$	0,0001047	0,0102299	0,0084086	0,1500788
ANFIS $DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0003164	0,0177888	0,0162739	0,2925936
Weighted average using $RMSE$	0,0000356	0,0059696	0,0051134	0,0919160

Source: Prepared by the author

Synthetic time series, generated using equation 13 with random seed 369. Univariate analysis identified the ARIMA$(1, 1, 1)$ model; based on this, tables 27 and 28 present the results for ANFIS modeling.

Table 27: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 369, with 1 lag

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,0015038	0,0387794	0,0332153	0,6321471	86,30
$DB(t-1)+Res$	0,0554476	0,2354732	0,1715343	3,2563966	143,83
$DB(t-1)+Res(t-1)$	0,0914559	0,3024168	0,1624823	3,0782162	163,71
$DB(t-1)+Dif$	0,0000188	0,0043411	0,0038710	0,0733735	495,42
$DB(t-1)+Dif(t-1)$	0,0057078	0,0755020	0,0677112	1,2868609	197,36
$DB(t-1)+Res+Dif$	0,0012826	0,0358135	0,0207042	0,3958795	311,93
$DB(t-1)+Res(t-1)+Dif$	0,0373161	0,1931737	0,0839085	1,5915088	355,42
$DB(t-1)+Res+Dif(t-1)$	0,0470088	0,2168151	0,1528790	2,9174146	228,22
$DB(t-1)+Res(t-1)+Dif(t-1)$	0,0198692	0,1409582	0,0980169	1,8542876	201,14

Source: Prepared by the author

Table 28: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 369, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,0151358	0,1230277	0,1159078	2,2005574	259,31
$DB(t-1)+DB(t-2)+Res$	0,0449091	0,2119176	0,1720879	3,2722948	197,15
$DB(t-1)+DB(t-2)+Res(t-1)$	0,0166879	0,1291815	0,1192085	2,2620477	222,85
$DB(t-1)+DB(t-2)+Dif$	0,0012762	0,0357245	0,0306358	0,5827811	311,65
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0038891	0,0623630	0,0582274	1,1027534	155,65
$DB(t-1)+DB(t-2)+Res+Dif$	0,2251529	0,4745028	0,2936811	5,6011581	610,36
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0336397	0,1834114	0,0910552	1,7247261	490,79
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	0,0598903	0,2447250	0,1829760	3,4845120	259,89
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,7445563	0,8628768	0,5078360	9,6100727	461,09

Source: Prepared by the author

Tables 29 and 30 present the results for the ensemble of selected models and a comparison of the best models, respectively.

Table 29: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 369

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,001119265	0,033455423	0,031966386	0,606596
Weighted average $(60\%ARIMA)$ $(40\%ANFIS)$	0,001667932	0,040840325	0,039133880	0,742590
Weighted average $(40\%ARIMA)$ $(60\%ANFIS)$	0,000680126	0,026079228	0,024798891	0,470602
Weighted average $(80\%ARIMA)$ $(20\%ANFIS)$	0,003093849	0,055622378	0,053468869	1,014578
Weighted average $(20\%ARIMA)$ $(80\%ANFIS)$	0,000130431	0,011420627	0,010463903	0,198614
Weighted average using $RMSE$	0,000006109	0,002471676	0,002344951	0,044539

Source: Prepared by the author

Table 30: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 369

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(1,1,1)$	0,0049579	0,0704123	0,0678040	1,2689450
ANFIS $DB(t-1)+Dif$	0,0000188	0,0043411	0,0038710	0,0733735
ANFIS $DB(t-1)+DB(t-2)+Dif$	0,0012762	0,0357245	0,0306358	0,5827811
Weighted average using $RMSE$	0,0014929	0,0386378	0,0339247	0,645998

Source: Prepared by the author

Synthetic time series, generated using equation 13, with random seed 580. Univariate analysis identified an ARIMA$(1,1,0)$ model; based on this, tables 31 and 32 present the results for ANFIS modeling.

Table 31: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 580, with 1 lag

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,001709	0,0413401	0,0295763	0,4027566	132,54
$DB(t-1)+Res$	0,0245995	0,1568422	0,1222693	1,6480352	229,22
$DB(t-1)+Res(t-1)$	0,0393425	0,1983495	0,1206344	1,6314022	151,42
$DB(t-1)+Dif$	0,0000533	0,0073018	0,0066393	0,0892674	404,26
$DB(t-1)+Dif(t-1)$	0,0026466	0,0514456	0,0427563	0,5800542	180,04
$DB(t-1)+Res+Dif$	0,0299630	0,1730982	0,1372508	1,8510591	412,32
$DB(t-1)+Res(t-1)+Dif$	0,0004921	0,0221840	0,0108023	0,1459364	2616,94
$DB(t-1)+Res+Dif(t-1)$	0,4990691	0,7064482	0,4354281	5,9072660	187,25
$DB(t-1)+Res(t-1)+Dif(t-1)$	0,0269215	0,1640777	0,1137356	1,5350383	202,54

Source: Prepared by the author

Table 32: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 580, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,004097	0,0640081	0,0584866	0,7915287	315,87
$DB(t-1)+DB(t-2)+Res$	0,3166523	0,5627187	0,3650690	4,9292489	274,16
$DB(t-1)+DB(t-2)+Res(t-1)$	0,0534837	0,2312655	0,2304805	3,1097556	274,76
$DB(t-1)+DB(t-2)+Dif$	0,0015398	0,0392403	0,0367806	0,4954830	1288,26
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0033160	0,0575846	0,0478231	0,6473053	201,29
$DB(t-1)+DB(t-2)+Res+Dif$	0,0490918	0,2215668	0,1572116	2,1345532	1198,02
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0569408	0,2386228	0,1817605	2,4572726	245,50
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	0,1854689	0,4306610	0,3207200	4,3133452	320,85
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,2064746	0,4543948	0,4289182	5,7910065	312,13

Source: Prepared by the author

Tables 33 and 34 present the results for the ensemble of selected models and a comparison of the best models, respectively.

Table 33: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 580

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,001248718	0,035337199	0,031117907	0,420501
Weighted average (60%$ARIMA$) (40%$ANFIS$)	0,001793140	0,042345482	0,037387069	0,505096
Weighted average (40%$ARIMA$) (60%$ANFIS$)	0,000806076	0,028391476	0,024848746	0,335906
Weighted average (80%$ARIMA$) (20%$ANFIS$)	0,003187326	0,056456405	0,049925391	0,674286
Weighted average (20%$ARIMA$) (80%$ANFIS$)	0,000226134	0,015037747	0,012976949	0,175669
Weighted average using $RMSE$	0,000083587	0,009142601	0,007634923	0,103458

Source: Prepared by the author

Table 34: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 580

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(1, 1, 0)$	0,0049886	0,0706302	0,0624636	0,846894
ANFIS $DB(t-1) + Dif$	0,0009764	0,0312477	0,0285739	0,386935
ANFIS $DB(t-1)+DB(t-2)+Dif$	0,0015398	0,0392403	0,0367806	0,495483
Weighted average using $RMSE$	0,0000523	0,0072289	0,0067475	0,090813

Source: Prepared by the author

Synthetic time series, generated using equation 14, with random seed number 62. Univariate analysis identified an ARIMA$(1, 1, 1)$ model; based on this, tables 35 and 36 present the results for ANFIS modeling.

Table 35: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 62, with 1 lag

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,0022166	0,0470809	0,0406355	0,5730268	109,90
$DB(t-1) + Res$	0,0125496	0,1120250	0,0951514	1,3361951	145,18
$DB(t-1) + Res(t-1)$	0,0079136	0,0889584	0,0744303	1,0478222	173,56
$DB(t-1) + Dif$	0,0000532	0,0072924	0,0067788	0,0951761	649,70
$DB(t-1) + Dif(t-1)$	0,0024760	0,0497595	0,0399823	0,5651257	170,93
$DB(t-1) + Res + Dif$	0,0104840	0,1023913	0,0670434	0,9387088	668,29
$DB(t-1)+Res(t-1)+Dif$	0,0000737	0,0085851	0,0057154	0,0801388	1566,42
$DB(t-1) + Res + Dif(t-1)$	0,1024384	0,3200601	0,2109979	2,9438085	295,91
$DB(t-1)+Res(t-1)+Dif(t-1)$	0,0680509	0,2608657	0,2038033	2,8625441	335,23

Source: Prepared by the author

Table 36: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 62, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,0019774	0,0444678	0,0335532	0,4735721	162,88
$DB(t-1)+DB(t-2)+Res$	0,3822188	0,6182385	0,3575474	4,9919085	187,71
$DB(t-1)+DB(t-2)+Res(t-1)$	0,0148899	0,1220241	0,1132534	1,5975183	367,04
$DB(t-1)+DB(t-2)+Dif$	0,0011705	0,0342132	0,0311183	0,4385153	1216,96
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0015442	0,0392966	0,0356194	0,5018021	339,98
$DB(t-1)+DB(t-2)+Res+Dif$	0,1247366	0,3531807	0,1842848	2,5718999	1700,60
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0121569	0,1102584	0,0830014	1,1635443	468,35
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	0,2462942	0,4962804	0,2750320	3,8344312	343,45
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,0492745	0,2219787	0,1872115	2,6284941	345,13

Source: Prepared by the author

Tables 37 and 38 present the results for the ensemble of selected models and a comparison of the best models, respectively.

Table 37: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 62

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,000674296	0,025967219	0,021507714	0,302118
Weighted average ($60\%ARIMA$) ($40\%ANFIS$)	0,001038056	0,032218876	0,026690429	0,374884
Weighted average ($40\%ARIMA$) ($60\%ANFIS$)	0,000390383	0,01975810	0,0167558690	0,235400
Weighted average ($80\%ARIMA$) ($20\%ANFIS$)	0,002005112	0,044778474	0,037055857	0,520415
Weighted average ($20\%ARIMA$) ($80\%ANFIS$)	0,000062091	0,007879813	0,007362634	0,103512
Weighted average using $RMSE$	0,000018935	0,004351387	0,003265907	0,045991

Source: Prepared by the author

Table 38: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 62

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(1,1,1)$	0,00329156	0,05737213	0,04742151	0,6599054
ANFIS $DB(t-1)+Dif$	0,00005320	0,00729240	0,00677880	0,0951761
ANFIS $DB(t-1)+DB(t-2)+Dif$	0,00117050	0,03421320	0,03111830	0,4385153
Weighted average using $RMSE$	0,00001894	0,00435139	0,00326591	0,0459910

Source: Prepared by the author

Synthetic time series, generated using equation 14 with random seed number 94. Univariate analysis identified an ARIMA$(2,1,1)$ model; based on this, tables 39, 40, and 41 present the results for the ANFIS modeling.

Table 39: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 94, with 1 lag

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,0003418	0,0184875	0,0133887	0,2181089	90,88
$DB(t-1)+Res$	0,0103817	0,1018906	0,0887181	1,4475392	142,26
$DB(t-1)+Res(t-1)$	0,0064910	0,0805666	0,0580705	0,9496992	83,81
$DB(t-1)+Dif$	0,0000256	0,0050553	0,0047341	0,0771229	464,02
$DB(t-1)+Dif(t-1)$	0,0013891	0,0372708	0,0309591	0,5059487	122,91
$DB(t-1)+Res+Dif$	0,0006789	0,0260567	0,0249575	0,4068821	294,36
$DB(t-1)+Res(t-1)+Dif$	0,0006570	0,0256325	0,0239615	0,3910587	230,44
$DB(t-1)+Res+Dif(t-1)$	0,0079003	0,0888835	0,0747516	1,2220416	228,07
$DB(t-1)+Res(t-1)+Dif(t-1)$	0,0058834	0,0767036	0,0593003	0,9693578	149,37

Source: Prepared by the author

Table 40: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 94, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,0018796	0,0433542	0,0393972	0,6426359	169,90
$DB(t-1)+DB(t-2)+Res$	0,0028580	0,0534603	0,0492785	0,8039795	186,70
$DB(t-1)+DB(t-2)+Res(t-1)$	0,0025731	0,0507260	0,0423227	0,691177	155,11
$DB(t-1)+DB(t-2)+Dif$	0,0003592	0,0189514	0,0130397	0,2131515	253,01
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0018084	0,0425252	0,0277350	0,4533328	254,21
$DB(t-1)+DB(t-2)+Res+Dif$	0,0014137	0,0375994	0,0254450	0,4155842	1256,79
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0086213	0,0928508	0,0691876	1,1309295	467,03
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	0,0103911	0,1019367	0,0662876	1,0816916	363,08
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,0097987	0,0989882	0,0695390	1,1372091	198,82

Source: Prepared by the author

Table 41: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 94, with 3 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1) + DB(t-2) + DB(t-3)$	0,0006577	0,0256462	0,0211629	0,3455111	218,34
$DB(t-1) + DB(t-2) + DB(t-3) + Res$	0,0074335	0,0862180	0,0590514	0,9638677	378,36
$DB(t-1) + DB(t-2) + DB(t-3) + Res(t-1)$	0,0342262	0,1850034	0,1148280	1,8793259	216,46
$DB(t-1) + DB(t-2) + DB(t-3) + Dif$	0,0028079	0,0529893	0,0497157	0,8106994	782,47
$DB(t-1) + DB(t-2) + DB(t-3) + Dif(t-1)$	0,0195866	0,1399520	0,1141868	1,8644603	302,35
$DB(t-1) + DB(t-2) + DB(t-3) + Res + Dif$	0,0132639	0,1151689	0,0673047	1,0992356	589,48
$DB(t-1) + DB(t-2) + DB(t-3)+Res(t-1)+Dif$	0,0039374	0,0627491	0,0559004	0,9121983	453,01
$DB(t-1) + DB(t-2) + DB(t-3)+Res+Dif(t-1)$	0,0084661	0,0920114	0,0763877	1,2426392	692,86
$DB(t-1) + DB(t-2) + DB(t-3) + Res(t-1) + Dif(t-1)$	0,0734865	0,2710840	0,1853044	3,0292883	390,99

Source: Prepared by the author

Tables 42 and 43 present the results for the ensemble of selected models and a comparison of the best models, respectively.

Table 42: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 94

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,000508659	0,022553470	0,018605943	0,303995
Weighted average (60%$ARIMA$) (40%$ANFIS$)	0,000696182	0,026385254	0,021594069	0,352847
Weighted average (40%$ARIMA$) (60%$ANFIS$)	0,000696182	0,026385254	0,021594069	0,352847
Weighted average (80%$ARIMA$) (20%$ANFIS$)	0,001162128	0,034090005	0,027658646	0,451979
Weighted average (20%$ARIMA$) (80%$ANFIS$)	0,000127894	0,011309042	0,010053851	0,164146
Weighted average using $RMSE$	0,000065676	0,008104102	0,007602293	0,124042

Source: Prepared by the author

Table 43: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 94

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(2, 1, 1)$	0,00174928	0,04182441	0,03427198	0,5646453
ANFIS $DB(t-1) + Dif$	0,00002560	0,00505530	0,00473410	0,0771229
ANFIS $DB(t-1)+DB(t-2)+Dif$	0,00035920	0,01895140	0,01303970	0,2131515
ANFIS $DB(t-1) + DB(t-2) + DB(t-3)$	0,00065770	0,02564620	0,02116290	0,3455111
Weighted average using $RMSE$	0,00006568	0,00810410	0,00760229	0,1240420

Source: Prepared by the author

Synthetic time series, generated using equation 14 with random seed 95. Univariate analysis identified an ARIMA$(1, 1, 1)$ model; based on this, tables 44 and 45 present the results for the ANFIS modeling.

Table 44: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 95, with 1 lag

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,0016213	0,0402649	0,0365572	0,4334834	157,39
$DB(t-1) + Res$	0,0045231	0,0672536	0,0565909	0,6687871	190,31
$DB(t-1) + Res(t-1)$	0,0292604	0,1710568	0,0956068	1,1233272	318,83
$DB(t-1) + Dif$	0,0000715	0,0084587	0,0081561	0,0965294	1059,03
$DB(t-1) + Dif(t-1)$	0,0017855	0,0422551	0,0352494	0,4191128	137,06
$DB(t-1) + Res + Dif$	0,2200750	0,4691216	0,2024684	2,3672301	1196,30
$DB(t-1)+Res(t-1)+Dif$	0,0049784	0,0705577	0,0289570	0,3386519	920,91
$DB(t-1) + Res + Dif(t-1)$	6,4627848	2,5422008	1,3815103	16,2155333	602,02
$DB(t-1)+Res(t-1)+Dif(t-1)$	1,7054991	1,3059476	0,5345321	6,2506496	535,35

Source: Prepared by the author

Table 45: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 95, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,001029	0,0320784	0,0283481	0,3373503	200,53
$DB(t-1)+DB(t-2)+Res$	0,0062112	0,0788114	0,0667981	0,7908562	209,98
$DB(t-1)+DB(t-2)+Res(t-1)$	0,0188220	0,1371935	0,1174464	1,3884458	242,42
$DB(t-1)+DB(t-2)+Dif$	0,0020126	0,0448618	0,0281901	0,3339440	354,40
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0046591	0,0682573	0,0671873	0,7976092	224,24
$DB(t-1)+DB(t-2)+Res+Dif$	0,0792948	0,2815933	0,1610582	1,8892541	1016,72
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0182465	0,1350795	0,0797432	0,9374000	1892,19
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	2,3961775	1,5479592	0,7169270	8,3897817	767,22
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,0639108	0,2528058	0,1441880	1,6939845	824,79

Source: Prepared by the author

Tables 46 and 47 present the results for the ensemble of selected models and a comparison of the best models, respectively.

Table 46: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 95

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,001062851	0,03260397	0,027698429	0,326974
Weighted average $(60\%ARIMA)$ $(40\%ANFIS)$	0,001619742	0,040246021	0,033869714	0,399722
Weighted average $(40\%ARIMA)$ $(60\%ANFIS)$	0,000625503	0,025010067	0,021527143	0,254227
Weighted average $(80\%ARIMA)$ $(20\%ANFIS)$	0,003092155	0,055607150	0,046212286	0,545218
Weighted average $(20\%ARIMA)$ $(80\%ANFIS)$	0,000109439	0,010461303	0,009184571	0,108731
Weighted average using $RMSE$	0,000032189	0,005673552	0,005071795	0,060344

Source: Prepared by the author

Table 47: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 95

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(1,1,1)$	0,005042738	0,07101224	0,05855481	0,6846114
ANFIS $DB(t-1)+Dif$	0,0000715	0,0084587	0,0081561	0,0965294
ANFIS $DB(t-1)+DB(t-2)$	0,0010290	0,0320784	0,0283481	0,3373503
Weighted average using $RMSE$	0,0000322	0,0056736	0,0050718	0,0603440

Source: Prepared by the author

Synthetic time series, generated using equation 14, with random seed 125. Univariate analysis identified an ARIMA$(1,1,1)$ model; based on this, tables 48 and 49 present the results for ANFIS modeling.

Table 48: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 125, with 1 lag

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,0012243	0,0349895	0,0309507	0,5373035	241,49
$DB(t-1)+Res$	5,4622092	2,3371370	2,0828083	36,1788167	622,79
$DB(t-1)+Res(t-1)$	0,5386184	0,7339063	0,5324306	9,2502642	1349,64
$DB(t-1)+Dif$	0,0002803	0,0167424	0,0165757	0,2878062	405,88
$DB(t-1)+Dif(t-1)$	0,0133291	0,1154516	0,1099537	1,9087933	145,79
$DB(t-1)+Res+Dif$	1,1624876	1,0781872	0,9257324	16,0772298	589,74
$DB(t-1)+Res(t-1)+Dif$	0,0151547	0,1231046	0,1021743	1,7747526	1315,47
$DB(t-1)+Res+Dif(t-1)$	0,3222895	0,5677054	0,4975790	8,6435452	555,90
$DB(t-1)+Res(t-1)+Dif(t-1)$	0,9349027	0,9669036	0,7936572	13,7869094	955,68

Source: Prepared by the author

Table 49: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 125, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,0180685	0,1344191	0,1248366	2,1687196	165,70
$DB(t-1)+DB(t-2)+Res$	0,1531608	0,3913576	0,3678626	6,3879892	578,56
$DB(t-1)+DB(t-2)+Res(t-1)$	0,3768587	0,6138882	0,5175984	8,9908138	541,70
$DB(t-1)+DB(t-2)+Dif$	0,0010169	0,0318893	0,0231720	0,4025286	244,80
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0110938	0,1053272	0,1018783	1,7694899	204,06
$DB(t-1)+DB(t-2)+Res+Dif$	0,0105480	0,1027035	0,0567485	0,9868079	1806,02
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0159002	0,1260959	0,1025107	1,7814903	1054,14
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	0,8011941	0,8950944	0,7828108	13,5997197	434,56
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,3828379	0,6187390	0,5234616	9,0972202	453,90

Source: Prepared by the author

Tables 50 and 51 present the results for the ensemble of selected models and a comparison of the best models, respectively.

Table 50: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 125

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,016304233	0,127688029	0,106246729	1,846993
Weighted average (60%$ARIMA$) (40%$ANFIS$)	0,024359384	0,156074930	0,130138669	2,262326
Weighted average (40%$ARIMA$) (60%$ANFIS$)	0,009865870	0,099327088	0,082354789	1,431659
Weighted average (80%$ARIMA$) (20%$ANFIS$)	0,045320051	0,212885065	0,177922549	3,092993
Weighted average (20%$ARIMA$) (80%$ANFIS$)	0,001839511	0,042889518	0,034570909	0,600992
Weighted average using $RMSE$	0,000067145	0,008194231	0,006949817	0,120673

Source: Prepared by the author

Table 51: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 125

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(1, 1, 1)$	0,0727480	0,269718	0,225707	3,7166650
ANFIS $DB(t-1) + Dif$	0,0002803	0,0167424	0,0165757	0,2878062
ANFIS $DB(t-1)+DB(t-2)+Dif$	0,0010169	0,0318893	0,0231720	0,4025286
Weighted average using $RMSE$	0,0000672	0,0081942	0,0069498	0,1206730

Source: Prepared by the author

Synthetic time series, generated using equation 14 with random seed 155. Univariate analysis identified the ARIMA$(1, 1, 1)$ model; based on this, tables 52 and 53 present the results for ANFIS modeling.

Table 52: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 155, with 1 lag

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,0021741	0,0466271	0,0449650	0,8266690	91,80
$DB(t-1) + Res$	0,0024052	0,0490427	0,0470337	0,8684499	251,13
$DB(t-1) + Res(t-1)$	0,0073138	0,0855206	0,0815256	1,5020776	124,21
$DB(t-1) + Dif$	0,0000186	0,0043180	0,0040554	0,0742388	341,97
$DB(t-1) + Dif(t-1)$	0,0036701	0,0605814	0,0576344	1,0597829	133,72
$DB(t-1) + Res + Dif$	0,0001184	0,0108804	0,0098889	0,1806609	520,20
$DB(t-1)+Res(t-1)+Dif$	0,0002495	0,0157949	0,0151308	0,2781477	465,52
$DB(t-1) + Res + Dif(t-1)$	0,0059395	0,0770680	0,0548154	1,0079859	190,22
$DB(t-1)+Res(t-1)+Dif(t-1)$	0,0016102	0,0401274	0,0353998	0,6549931	181,05

Source: Prepared by the author

Table 53: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 155, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,0033266	0,057677	0,0522151	0,9614777	138,07
$DB(t-1)+DB(t-2)+Res$	0,0027928	0,0528468	0,0500921	0,9188126	194,73
$DB(t-1)+DB(t-2)+Res(t-1)$	0,0107795	0,1038243	0,0945629	1,7301897	308,26
$DB(t-1)+DB(t-2)+Dif$	0,0003377	0,0183779	0,0167779	0,3086199	352,89
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0012405	0,0352204	0,0326384	0,6026030	170,85
$DB(t-1)+DB(t-2)+Res+Dif$	0,0002383	0,0154380	0,0106349	0,1982028	627,53
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0007910	0,0281239	0,0190540	0,3544467	484,65
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	0,0054298	0,0736868	0,0664319	1,2273679	263,36
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,0009999	0,0316217	0,0269504	0,4960036	246,30

Source: Prepared by the author

Tables 54 and 55 present the results for the ensemble of selected models and a comparison of the best models, respectively.

Table 54: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 155

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,000109189	0,010449354	0,007588021	0,141649
Weighted average ($60\%ARIMA$) ($40\%ANFIS$)	0,000167770	0,012952588	0,009454589	0,176348
Weighted average ($40\%ARIMA$) ($60\%ANFIS$)	0,000064099	0,008006196	0,005772529	0,107890
Weighted average ($80\%ARIMA$) ($20\%ANFIS$)	0,000325403	0,018038920	0,013247134	0,246828
Weighted average ($20\%ARIMA$) ($80\%ANFIS$)	0,000014392	0,003793624	0,003209017	0,059731
Weighted average using $RMSE$	0,000010348	0,003216850	0,003075959	0,056836

Source: Prepared by the author

Table 55: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 155

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(1,1,1)$	0,00053700	0,0231732	0,01733925	0,3208939
ANFIS $DB(t-1)+Dif$	0,00001860	0,0043180	0,00405540	0,0742388
ANFIS $DB(t-1)+DB(t-2)+Res+Dif$	0,00023830	0,0154380	0,01063490	0,1982028
Weighted average using $RMSE$	0,00001035	0,0032169	0,00307596	0,0568360

Source: Prepared by the author

Synthetic time series, generated using equation 14 with random seed 185. Univariate analysis identified the

ARIMA$(2, 1, 1)$ model; based on this, tables 56, 57, and 58 present the results for the ANFIS modeling.

Table 56: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 185, with 1 lag

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,0023039	0,0479992	0,0395947	0,7315498	259,03
$DB(t-1)+Res$	0,0160691	0,1267641	0,1081565	2,0021241	892,22
$DB(t-1)+Res(t-1)$	0,0086418	0,0929613	0,0796093	1,4581683	654,52
$DB(t-1)+Dif$	0,0000454	0,0067393	0,0061902	0,1135229	882,53
$DB(t-1)+Dif(t-1)$	0,0033849	0,0581800	0,0468807	0,8650579	385,91
$DB(t-1)+Res+Dif$	0,0223594	0,1495305	0,0716869	1,3389591	934,88
$DB(t-1)+Res(t-1)+Dif$	0,0008580	0,0292909	0,0224919	0,4170529	941,49
$DB(t-1)+Res+Dif(t-1)$	0,0037469	0,0612120	0,0500178	0,9224254	608,85
$DB(t-1)+Res(t-1)+Dif(t-1)$	0,8872530	0,9419411	0,4754176	8,8094046	518,10

Source: Prepared by the author

Table 57: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 185, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,0067695	0,0822769	0,0708416	1,3125001	382,28
$DB(t-1)+DB(t-2)+Res$	0,0570269	0,2388031	0,1226291	2,2896859	658,89
$DB(t-1)+DB(t-2)+Res(t-1)$	0,1193069	0,3454082	0,2448162	4,5452763	537,19
$DB(t-1)+DB(t-2)+Dif$	0,0003670	0,0191575	0,0174061	0,3212737	921,05
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0353248	0,1879489	0,1398390	2,5847280	1423,21
$DB(t-1)+DB(t-2)+Res+Dif$	0,0004184	0,0204559	0,0160820	0,2981771	982,14
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0582864	0,2414258	0,1670781	3,1094916	1142,49
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	0,0191336	0,1383243	0,1221959	2,2368838	865,77
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,0569608	0,2386646	0,1965024	3,6358742	871,81

Source: Prepared by the author

Table 58: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 185, with 3 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)+DB(t-3)$	0,0187312	0,1368622	0,1246695	2,2859823	499,53
$DB(t-1)+DB(t-2)+DB(t-3)+Res$	0,0214850	0,1465778	0,1290845	2,3719910	682,16
$DB(t-1)+DB(t-2)+DB(t-3)+Res(t-1)$	0,0309741	0,1759945	0,1596903	2,9344531	679,04
$DB(t-1)+DB(t-2)+DB(t-3)+Dif$	0,0032804	0,0572752	0,0386938	0,7187499	836,04
$DB(t-1)+DB(t-2)+DB(t-3)+Dif(t-1)$	0,0071249	0,0844088	0,0748856	1,3753941	738,27
$DB(t-1)+DB(t-2)+DB(t-3)+Res+Dif$	0,0053078	0,0728543	0,0517599	0,9598953	2464,63
$DB(t-1)+DB(t-2)+DB(t-3)+Res(t-1)+Dif$	0,0184776	0,1359324	0,1016912	1,8845575	3861,19
$DB(t-1)+DB(t-2)+DB(t-3)+Res+Dif(t-1)$	0,0076447	0,0874339	0,0652294	1,2059017	1212,58
$DB(t-1)+DB(t-2)+DB(t-3)+Res(t-1)+Dif(t-1)$	0,3961460	0,6294013	0,5304674	9,8479673	2550,81

Source: Prepared by the author

Tables 59 and 60 present the results for the ensemble of selected models and a comparison of the best models, respectively.

Table 59: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 185

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,000806683	0,0284022	0,025008114	0,463242
Weighted average $(60\%ARIMA)$ $(40\%ANFIS)$	0,001172787	0,0342460	0,030706091	0,568408
Weighted average $(40\%ARIMA)$ $(60\%ANFIS)$	0,000511862	0,0226244	0,019450474	0,360624
Weighted average $(80\%ARIMA)$ $(20\%ANFIS)$	0,002118848	0,0460309	0,042102046	0,778739
Weighted average $(20\%ARIMA)$ $(80\%ANFIS)$	0,000136073	0,0116650	0,009355680	0,173909
Weighted average using $RMSE$	0,000057115	0,0075575	0,006062040	0,112417

Source: Prepared by the author

Table 60: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 185

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(2,1,1)$	0,00335005	0,0578796	0,0534980	0,9776734
ANFIS $DB(t-1)+Dif$	0,00004540	0,0067393	0,0061902	0,1135229
ANFIS $DB(t-1)+DB(t-2)+Dif$	0,00036700	0,0191575	0,0174061	0,3212737
ANFIS $DB(t-1)+DB(t-2)+DB(t-3)+Dif$	0,00328040	0,0572752	0,0386938	0,7187499
Weighted average using $RMSE$	0,00005712	0,0075575	0,0060620	0,1124170

Source: Prepared by the author

Synthetic time series, generated using equation 14, with a random seed of 250. Univariate analysis identified the ARIMA$(1, 1, 1)$ model; based on this, tables 61 and 62 present the results for the ANFIS modeling.

Table 61: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 250, with 1 lag

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,0005093	0,0225681	0,0180581	0,3069868	215,35
$DB(t-1)+Res$	0,0019706	0,0443918	0,0353590	0,6008770	217,67
$DB(t-1)+Res(t-1)$	0,0025251	0,0502506	0,0344721	0,5847054	403,93
$DB(t-1)+Dif$	0,0000139	0,0037299	0,0034924	0,0593624	621,82
$DB(t-1)+Dif(t-1)$	0,0004359	0,0208776	0,0174883	0,2981334	303,83
$DB(t-1)+Res+Dif$	0,1825123	0,4272146	0,2354838	3,9826566	333,72
$DB(t-1)+Res(t-1)+Dif$	0,0060616	0,0778565	0,0350821	0,5922428	883,27
$DB(t-1)+Res+Dif(t-1)$	0,0027192	0,0521460	0,0376361	0,6379468	306,17
$DB(t-1)+Res(t-1)+Dif(t-1)$	0,1339083	0,3659348	0,1698653	2,8688476	634,56

Source: Prepared by the author

Table 62: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 250, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,0007935	0,0281689	0,0215069	0,3651216	766,21
$DB(t-1)+DB(t-2)+Res$	0,0900009	0,3000014	0,1847239	3,1287847	341,09
$DB(t-1)+DB(t-2)+Res(t-1)$	0,0666068	0,2580829	0,1145384	1,9336512	679,91
$DB(t-1)+DB(t-2)+Dif$	0,0000677	0,0082255	0,0064012	0,1090565	905,63
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0024475	0,0494719	0,0454363	0,7727618	464,94
$DB(t-1)+DB(t-2)+Res+Dif$	0,0485459	0,2203313	0,1247670	2,1115997	1369,24
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0265063	0,1628074	0,0666325	1,1245334	2416,63
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	0,7374576	0,8587535	0,4817591	8,1500858	434,46
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,0927735	0,3045875	0,1557840	2,6324266	458,90

Source: Prepared by the author

Tables 63 and 64 present the results for the ensemble of selected models and a comparison of the best models, respectively.

Table 63: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 250

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,000756201	0,027499115	0,023710314	0,402642
Weighted average (60%$ARIMA$) (40%$ANFIS$)	0,001134648	0,033684535	0,029150823	0,495043
Weighted average (40%$ARIMA$) (60%$ANFIS$)	0,000454418	0,021317073	0,018269806	0,310242
Weighted average (80%$ARIMA$) (20%$ANFIS$)	0,002121530	0,046060065	0,040031840	0,679843
Weighted average (20%$ARIMA$) (80%$ANFIS$)	0,000080839	0,008991033	0,007388789	0,125441
Weighted average using $RMSE$	0,000001791	0,001338248	0,001016867	0,017255

Source: Prepared by the author

Table 64: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 250

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(1, 1, 1)$	0,00341504	0,05843838	0,05091275	0,8549588
ANFIS $DB(t-1) + Dif$	0,00001390	0,00372990	0,00349240	0,0593624
ANFIS $DB(t-1)+DB(t-2)+Dif$	0,00006770	0,00822550	0,00640120	0,1090565
Weighted average using $RMSE$	0,00000179	0,00133825	0,00101687	0,0172550

Source: Prepared by the author

Synthetic time series, generated using equation 14 with random seed 395. Univariate analysis identified the ARIMA$(1, 1, 1)$ model; based on this, tables 65 and 66 present the results for ANFIS modeling.

Table 65: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 395, with 1 lag

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)$	0,0010458	0,0323387	0,0253668	0,5127494	255,50
$DB(t-1) + Res$	0,0015263	0,0390674	0,0297789	0,6017811	255,48
$DB(t-1) + Res(t-1)$	0,0010614	0,0325787	0,0253360	0,5121664	403,52
$DB(t-1) + Dif$	0,0000022	0,0014814	0,0012060	0,0243947	511,76
$DB(t-1) + Dif(t-1)$	0,0018932	0,0435107	0,0364054	0,7359677	544,23
$DB(t-1) + Res + Dif$	0,0000010	0,0009935	0,0008001	0,0162091	492,60
$DB(t-1)+Res(t-1)+Dif$	0,0000022	0,0014892	0,0012651	0,0255989	796,36
$DB(t-1) + Res + Dif(t-1)$	0,0188921	0,1374484	0,0934654	1,8876890	388,49
$DB(t-1)+Res(t-1)+Dif(t-1)$	0,0020240	0,0449889	0,0397567	0,8039842	575,99

Source: Prepared by the author

Table 66: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ANFIS model using seed 395, with 2 lags

ANFIS Model	**MSE**	**RMSE**	**MAE**	**MAPE(%)**	**Execution time (s)**
$DB(t-1)+DB(t-2)$	0,0008995	0,0299919	0,0264991	0,5362394	686,30
$DB(t-1)+DB(t-2)+Res$	0,0017939	0,0423540	0,0369645	0,7475704	463,49
$DB(t-1)+DB(t-2)+Res(t-1)$	0,0012702	0,0356395	0,0311622	0,6299132	720,17
$DB(t-1)+DB(t-2)+Dif$	0,0000035	0,0018787	0,0015269	0,0308802	1659,20
$DB(t-1)+DB(t-2)+Dif(t-1)$	0,0016488	0,0406052	0,0336623	0,6808728	718,85
$DB(t-1)+DB(t-2)+Res+Dif$	0,0084175	0,0917470	0,0414885	0,8372442	1053,24
$DB(t-1)+DB(t-2)+Res(t-1)+Dif$	0,0000744	0,0086248	0,0061448	0,1245126	1087,24
$DB(t-1)+DB(t-2)+Res+Dif(t-1)$	0,0141662	0,1190217	0,0901922	1,8229830	631,51
$DB(t-1)+DB(t-2)+Res(t-1)+Dif(t-1)$	0,0055876	0,0747505	0,0627258	1,2682524	811,45

Source: Prepared by the author

Tables 67 and 68 present the results for the ensemble of selected models and a comparison of the best-performing models, respectively.

Table 67: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the ensemble models, using seed 395

Ensemble metric applied	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
Simple Average	0,000709825	0,026642550	0,021663029	0,437932
Weighted Average ($60\%ARIMA$) ($40\%ANFIS$)	0,001019564	0,031930613	0,026018594	0,525988
Weighted Average ($40\%ARIMA$) ($60\%ANFIS$)	0,000456077	0,021355964	0,017307463	0,349876
Weighted Average ($80\%ARIMA$) ($20\%ANFIS$)	0,001807012	0,042508965	0,034729726	0,702101
Weighted Average ($20\%ARIMA$) ($80\%ANFIS$)	0,000116552	0,010795902	0,008596331	0,173763
Weighted Average using the $RMSE$	0,000002261	0,001503640	0,001046063	0,021167

Source: Prepared by the author

Table 68: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) for the best models at each stage for seed 395

Metric	**MSE**	**RMSE**	**MAE**	**MAPE(%)**
ARIMA $(1,1,0)$	0,00281843	0,05308885	0,04344087	0,8743946
ANFIS $DB(t-1)+Res+Dif$	0,00000100	0,00099350	0,00080010	0,0162091
ANFIS $DB(t-1)+DB(t-2)+Dif$	0,00000350	0,00187870	0,00152690	0,0308802
Weighted Average using the $RMSE$	0,00000226	0,00150364	0,00104606	0,0211670

Source: Prepared by the author